# The eigenproblem

For $A \in \mathbb{C}^{n \times n}$, suppose a scalar $\lambda$ and nonzero vector $x$ satisfy

$$Ax = \lambda x,$$

Then $\lambda$ is an eigenvalue and $x$ an eigenvector.

Equivalently,

$$(A - \lambda I)x = 0.$$

Since $x \neq 0$, the matrix $A - \lambda I$ is singular. Hence

$$p_A(\lambda) = \det(\lambda I - A) = 0.$$

The one-dimensional subspace $\operatorname{span}\{x\}$ is invariant. Over the reals, a negative eigenvalue reverses the arrow direction while preserving the line.

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## Characteristic polynomial, trace, and determinant

Using the monic convention for an $n \times n$ matrix,

$$p_A(\lambda) = \lambda^n - a_1\lambda^{n-1} + \cdots + (-1)^n \det A.$$

Counting eigenvalues with algebraic multiplicity gives

$$\sum_i \lambda_i = \operatorname{tr} A, \qquad \prod_i \lambda_i = \det A.$$

For example,

$$A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}.$$

The trace is 6 and determinant is 8.

$$p_A(\lambda) = \lambda^2 - 6\lambda + 8 = (\lambda - 4)(\lambda - 2).$$

The eigenvalues are 4 and 2. Trace and determinant check the spectrum but do not generally determine it in dimensions above two.

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## How operations transform eigenvalues

If $Ax = \lambda x$, then for nonnegative integers $k$ and polynomials $q$,

$$A^k x = \lambda^k x,$$

$$A^{-1} x = \lambda^{-1} x \quad (A \text{ invertible}),$$

$$q(A)x = q(\lambda)x.$$

Shift:

$$(A + cI)x = (\lambda + c)x.$$

Eigenspace:

$$E_\lambda = N(A - \lambda I).$$

The eigenspace is a subspace whose dimension is the geometric multiplicity. A triangular matrix has its diagonal entries as eigenvalues, counted with algebraic multiplicity.

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## Algebraic and geometric multiplicities

The multiplicity of a root in the characteristic polynomial is its algebraic multiplicity $a_\lambda$. The quantity

$$g_\lambda = \dim N(A - \lambda I)$$

is its geometric multiplicity. Extend an eigenspace basis to a basis of the whole space. The resulting matrix is block upper triangular with block $\lambda I_{g_\lambda}$, so its characteristic polynomial contains $(t - \lambda)^{g_\lambda}$. Therefore

$$1 \leq g_\lambda \leq a_\lambda.$$

If any eigenvalue has $g_\lambda < a_\lambda$, the matrix is defective.

For example,

$$A = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}.$$

The algebraic multiplicity is 2. Solving

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# Algebraic and geometric multiplicities (continued)

$$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} x = 0$$

gives only $x_2 = 0$, hence

$$E_2 = \operatorname{span}\{(1,0)^T\}, \qquad g_2 = 1.$$

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# A complete two-dimensional example

$$A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}.$$

$\lambda = 4$ :

$$\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} x = 0,$$

Choose $x_1 = (1, 1)^T$.

$\lambda = 2$ :

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} x = 0,$$

Choose $x_2 = (1, -1)^T$.

These eigenvectors are orthogonal because $A$ is symmetric. If

$$u_0 = c_1 x_1 + c_2 x_2,$$

Then

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## A complete two-dimensional example (continued)

$$u_k = A^k u_0 = c_1 4^k x_1 + c_2 2^k x_2.$$

For $c_1 \neq 0$, the eigenvalue-4 mode determines the limiting direction.

Taking $c_1 = c_2 = 1$,

$$u_0 = x_1 + x_2 = (2, 0)^T, \quad u_1 = (6, 2)^T, \quad u_2 = (20, 12)^T.$$

The diagram shows $c_1 4^k x_1$, $c_2 2^k x_2$, and their sum $u_k$. For $c_1 = 0, c_2 = 1$, instead,

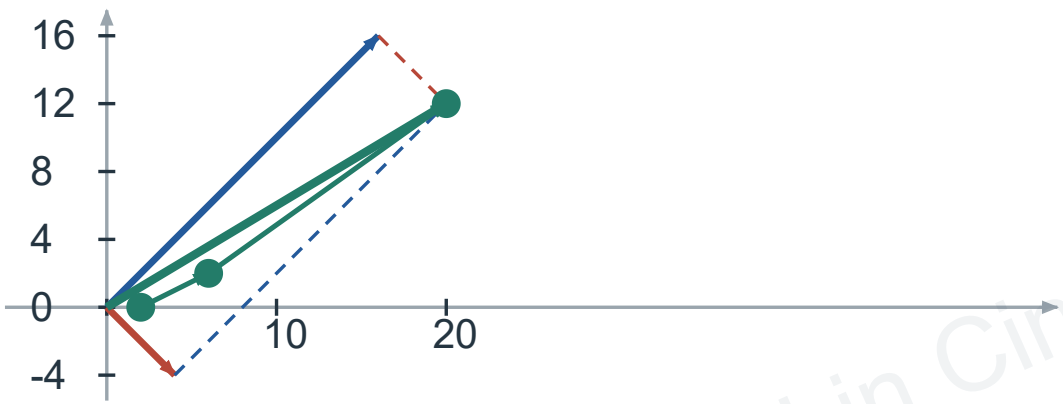
$$u_0 = (1, -1)^T, \quad u_1 = (2, -2)^T, \quad u_2 = (4, -4)^T,$$

the trajectory stays in the eigenvector direction $x_2$. It has no component in the larger-eigenvalue mode. Real coefficients can also be used with unit eigenvectors; here the diagram retains the specified nonunit $x_1, x_2$, so its coefficients refer to that basis.

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# A complete two-dimensional example (continued)

Eigenmodes: components, sum, and trajectory




**A=[[3,1],[1,3]]**

$\lambda_1=4 \cdot x_1=(1,1)$

$\lambda_2=2 \cdot x_2=(1,-1)$

**Blue: $4^2x_1=(16,16)$**

**Red: $2^2x_2=(4,-4)$**

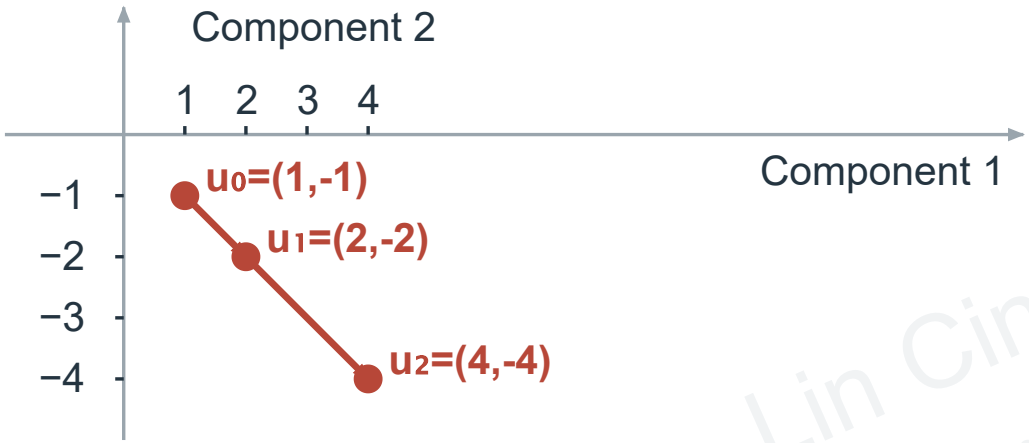
**Green: $u_2=(20,12)$**

$u_0,u_1,u_2=(2,0),(6,2),(20,12)$

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# A complete two-dimensional example (continued)

Exceptional trajectory $c_1=0$: always along $x_2=(1,-1)$




**$c_1=0$ · $c_2=1$**

$u_k=2^k(1,-1)$

Every $u_k$ has the same eigenvector direction.

**Fixed ratio; no $4^k$ dominant term.**

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## Rotation and complex eigenvalues

$$R_\theta = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}.$$

$$p(\lambda) = \lambda^2 - 2\cos\theta\,\lambda + 1,$$

$$\lambda = e^{\pm i\theta}.$$

Unless $\theta \equiv 0 \pmod{\pi}$, a planar rotation has no nonzero real eigenvector.

Complex eigenvalues of a real matrix come in conjugate pairs. If

$$Ax = \lambda x,$$

taking complex conjugates gives

$$A\overline{x} = \overline{\lambda}\,\overline{x}.$$

Real solutions require compatible conjugate coefficients. A single complex mode is generally not a real physical solution.

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# Diagonalization

Suppose $A$ has $n$ independent eigenvectors $x_i$. Form

$$X = [x_1 \ \cdots \ x_n].$$

$$AX = X\Lambda, \qquad \Lambda = \operatorname{diag}(\lambda_i),$$

$$A = X\Lambda X^{-1}, \qquad X^{-1}AX = \Lambda.$$

Diagonalization is equivalent to the eigenspace dimensions summing to $n$. Eigenvectors for distinct eigenvalues are independent. Indeed, a shortest nontrivial relation $\sum c_i x_i = 0$, after applying $A - \lambda_k I$, would give a shorter relation, a contradiction. Thus $n$ distinct eigenvalues suffice. Repeated eigenvalues can also have complete eigenspaces, as for $2I$.

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## Matrix powers and discrete-time stability

For $A = X\Lambda X^{-1}$, we have $A^k = X\Lambda^k X^{-1}$. If $u_0 = \sum_j c_j x_j$, the recurrence $u_{k+1} = Au_k$ has solution $u_k = \sum_j c_j \lambda_j^k x_j$. This eigenvector expansion requires diagonalizability.

The s×s Jordan block $J_s(\lambda) = \lambda I_s + N_s$ has λ on its diagonal, 1 directly above it, and 0 elsewhere; thus $N_s^s = 0$. Use these blocks for a general square matrix. If $J = \lambda I + N$, $N^m = 0$, and $\lambda \neq 0$, the binomial theorem gives

$$J^k = \sum_{j=0}^{\min(k,m-1)} \binom{k}{j} \lambda^{k-j} N^j.$$

When $0 < |\lambda| < 1$, each coefficient is a polynomial of degree at most j times geometric decay, hence tends to zero. When $\lambda = 0$, simply use $J^k = 0$ for $k \geq m$, without negative powers.

Thus $A^k \to 0$ exactly when $\rho(A) = \max_j |\lambda_j| < 1$. Necessity follows on an eigenvector: if $|\lambda| \geq 1$, then $A^k x = \lambda^k x$ does not tend to zero.

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# What else matters on the unit circle?

| Eigenvalues and Jordan structure | Long-term behavior of $A^k$ |
|---|---|
| All eigenvalue moduli below 1 | Tends to zero, even if defective |
| All moduli at most 1; unit-circle blocks have size 1 | Bounded; any unit-circle mode prevents decay to zero |
| A unit-circle block has size greater than 1 | Polynomial growth on some generalized eigenvectors |
| An eigenvalue has modulus greater than 1 | Exponential growth on its eigenvector |

For example I preserves the initial state, but $J = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ gives $J^k e_2 = (k, 1)^T$. Both have only eigenvalue 1; block size distinguishes them. Rotation powers are bounded, but an angle that is an irrational multiple of $2\pi$ has no common finite period.

These are long-term statements. Even $\rho(A) < 1$ permits finite-time growth; eigenvalues alone do not imply a decreasing Euclidean norm.

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## Similarity invariants

If

$$C = B^{-1}AB,$$

then

$$C - \lambda I = B^{-1}(A - \lambda I)B,$$

Thus characteristic polynomial, eigenvalues, trace, determinant, rank, and Jordan structure are unchanged.

If $Ax = \lambda x$, set $y = B^{-1}x$. Then

$$Cy = \lambda y.$$

Similarity changes coordinates while preserving the intrinsic spectrum.

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# Jordan chains supply the missing direction

Take $A = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix} = 2I + N$. The equation $(A - 2I)x = 0$ gives only $x_2 = 0$: its eigenspace has dimension 1, so A is not diagonalizable. Put $e_1 = (1,0)^T, e_2 = (0,1)^T$.

## The chain arrows apply N, not A

$$e_2 = (0,1)^T \xrightarrow{N} e_1 = (1,0)^T \xrightarrow{N} 0$$

$N^2 e_2 = 0$ but $N e_2 \neq 0$, so the chain has length 2.

**Apply A to $e_1$**

$Ae_1 = 2e_1$

Eigenvector

**Apply A to $e_2$**

$Ae_2 = e_1 + 2e_2$

Generalized eigenvector; image is not parallel

A=2I+N, N=[[0,1],[0,0]]. The basis ($e_1$,$e_2$) contains only one eigenvector.

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## Where does the polynomial factor in a Jordan power come from?

Since $N^2 = 0$ and $2I$ commutes with N, only two terms survive the binomial expansion. For integers $k \geq 0$,

$$A^k = 2^k I + k 2^{k-1} N = \begin{bmatrix} 2^k & k2^{k-1} \\ 0 & 2^k \end{bmatrix}.$$

Take $u_0 = e_2$ and write $u_k = (a_k, b_k)^T$. Matrix multiplication gives $a_{k+1} = 2a_k + b_k$, $b_{k+1} = 2b_k$, so $b_k = 2^k$ and

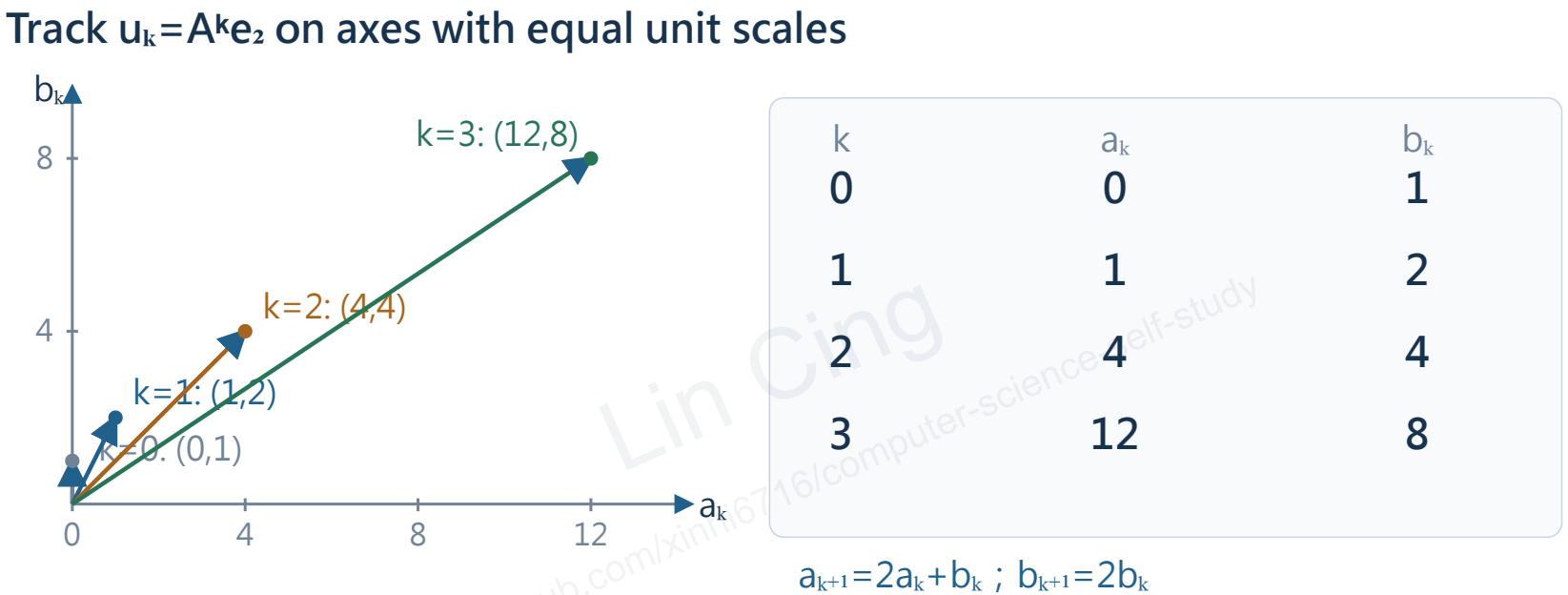
$$a_k = \sum_{j=0}^{k-1} 2^{k-1-j} b_j = \sum_{j=0}^{k-1} 2^{k-1} = k2^{k-1} \quad (k \geq 1).$$

At k=0 the empty sum gives $a_0 = 0$. Every step injects another $e_1$ component; k such contributions produce the polynomial factor.

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# Matrix multiplication and the closed form give the same trajectory

Use the same A with $u_0 = e_2$. Each arrow starts at the origin and ends at $A^k e_2$.

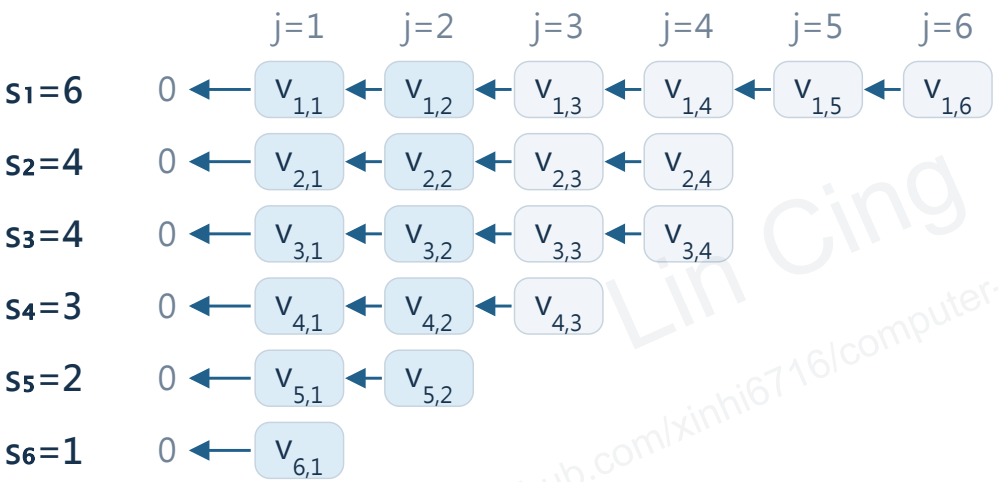



Check $A(4,4)^T = (12,8)^T$. Replacing A by 2I would give $(0, 2^k)^T$, losing the entire horizontal component.

# Recovering Jordan blocks from kernel dimensions

Let N be nilpotent, with chain lengths $s_1, \ldots, s_r$. Set $d_k = \dim \ker N^k$, $d_0 = 0$. Each chain contributes precisely its first $\min(k, s_i)$ vectors to the kernel of $N^k$.

### N sends each cell one position left; the first goes to 0




k=2

dim ker(N²)

=2+2+2+2+2+1

=11

First min(k,s) cells per chain lie in ker(Nᵏ).

Blue cells form a basis of ker(N²); each row is a separate Jordan block.

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## What does each nullity increment count?

A chain of length at least k contributes one new basis vector between $\ker N^{k-1}$ and $\ker N^k$. Therefore

$$d_k - d_{k-1} = \#\{i : s_i \geq k\}, \qquad \#\{i : s_i = k\} = 2d_k - d_{k-1} - d_{k+1}.$$

For $(d_0, \ldots, d_6) = (0, 6, 11, 15, 18, 19, 20)$, with $d_7$=20, the increments are (6,5,4,3,1,1,0). Adjacent differences give block counts (1,1,1,2,0,1) for lengths 1 through 6. Thus the lengths are 6,4,4,3,2,1, totaling 20 dimensions.

**Checks.** The value $d_1$=6 counts blocks. The first full nullity $d_6$=20 identifies maximum block length 6, so $N^6$=0 but $N^5 \neq 0$. For a general eigenvalue λ, apply the same method to N=A−λI on its generalized eigenspace.

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# Spectral decomposition

If the rows of $X^{-1}$ are $y_j^T$,

$$A = \sum_{j=1}^{n} \lambda_j x_j y_j^T.$$

Each $x_j y_j^T$ has rank one. For a real symmetric matrix choose a real orthogonal matrix $Q$:

$$A = Q \Lambda Q^T = \sum_j \lambda_j q_j q_j^T.$$

This scales independent orthonormal directions. Nonnormal matrices generally have nonorthogonal eigenvectors, so eigenvalues alone do not describe transient amplification.

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## From scalar to matrix ODEs

Scalar:

$$u' = \lambda u \quad \Longrightarrow \quad u(t) = Ce^{\lambda t}.$$

System:

$$u' = Au, \qquad u(0) = u_0.$$

If $A$ is diagonalizable,

$$u(t) = \sum_j c_j e^{\lambda_j t} x_j.$$

$$c = X^{-1} u_0.$$

For $\lambda = \alpha + i\beta$,

$$e^{\lambda t} = e^{\alpha t}(\cos \beta t + i \sin \beta t).$$

The real part determines exponential growth or decay. The imaginary part determines oscillation.

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## The matrix exponential

$$e^{At} = I + At + \frac{A^2t^2}{2!} + \cdots.$$

The absolutely convergent power series can be differentiated term by term on bounded time intervals.

$$\frac{d}{dt}e^{At} = Ae^{At} = e^{At}A.$$

Diagonalizable:

$$e^{At} = Xe^{\Lambda t}X^{-1} = \sum_j e^{\lambda_j t}x_j y_j^T.$$

Defective $A = 2I + N$, $N^2 = 0$:

$$e^{At} = e^{2t}(I + tN).$$

Eigenvalue real parts determine asymptotic stability. Nonnormal transient behavior also depends on norms or more detailed information about the matrix.

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## Stability and its boundary cases

$$\begin{bmatrix} u_1' \\ u_2' \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}.$$

Use eigenbasis：

$$u(t) = c_1 e^{4t}(1,1)^T + c_2 e^{2t}(1,-1)^T.$$

Both eigenvalues are positive, so every nonzero initial vector produces a growing solution.

The zero solution of a linear system is asymptotically stable exactly when every eigenvalue has negative real part. A Jordan block $J = \lambda I + N$ of size $m$ gives $e^{Jt} = e^{\lambda t} \sum_{\ell=0}^{m-1} t^\ell N^\ell / \ell!$. A negative real part makes every polynomial times the decaying exponential tend to zero. A positive real part gives an unstable mode.

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## Stability and its boundary cases (continued)

At the boundary, if all real parts are nonpositive and every imaginary-axis Jordan block has size one, $e^{At}$ is bounded and the zero solution is Lyapunov stable. Any remaining imaginary-axis mode prevents asymptotic stability. For $A = 0$, solutions retain their initial values; for skew-symmetric rotation, they retain their length. In contrast, $J = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ gives $e^{Jt} = I + tJ$, taking $e_2$ to $(t, 1)^T$. Zero real parts can therefore still be unstable. These conclusions concern the stated linear system.

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# Asymptotic stability still permits initial norm growth

Let $B = \begin{bmatrix} -1 & 4 \\ 0 & -1 \end{bmatrix}$, $u_0 = (1,1)^T/\sqrt{2}$. Both eigenvalues are −1, but

$$u(t) = e^{-t}(1+4t,1)^T/\sqrt{2}, \qquad g(t) = \|u(t)\|_2 = e^{-t}\sqrt{\frac{(1+4t)^2+1}{2}}.$$

Put $h(t) = g(t)^2$. Differentiation gives

$$h'(t) = e^{-2t}\{4(1+4t) - (1+4t)^2 - 1\}, \qquad h'(0) = 2 > 0.$$

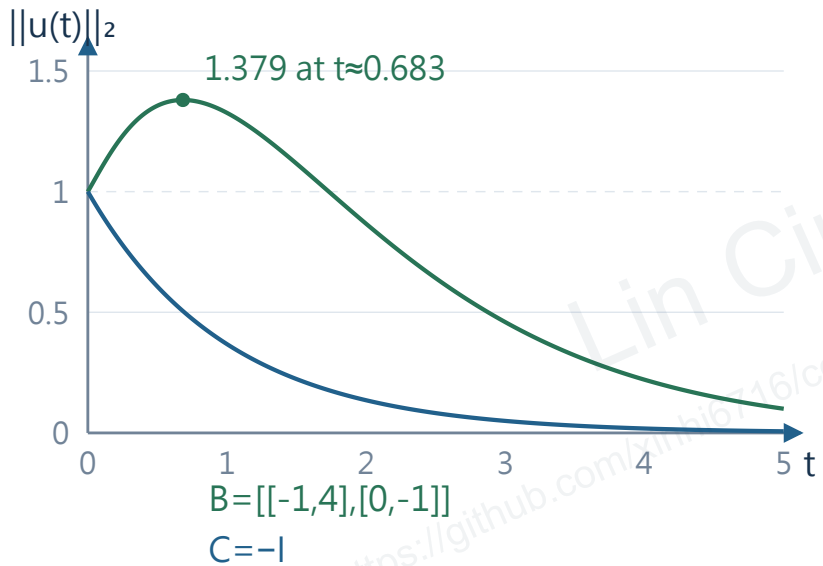
Thus this unit initial state grows first. With z=1+4t, stationarity gives $z^2-4z+1=0$. Only $z=2+\sqrt{3}$ corresponds to $t \geq 0$, so $t_* = (1+\sqrt{3})/4 \approx 0.683$, $g(t_*) \approx 1.379$. All components eventually decay because $te^{-t} \to 0$.

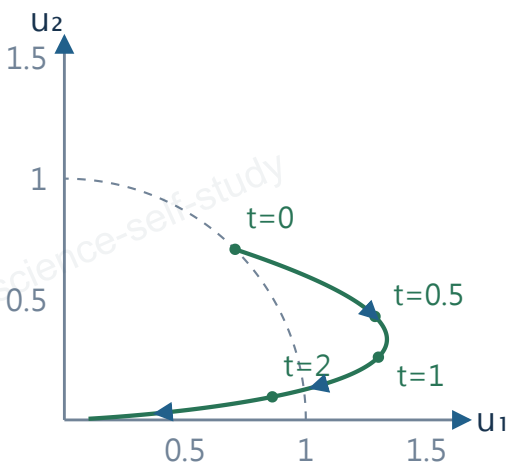
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# Why can the same eigenvalues give different curves?

Compare B with C=−I, using the same unit initial state $u_0 = (1,1)^T/\sqrt{2}$. For C the norm is exactly $e^{-t}$. For B the off-diagonal term feeds the second component into the first.

**Same eigenvalues −1,−1 and unit initial state; different transient norms**




B trajectory; dashed unit circle

Equal unit scales on both axes.

The peak is for the **specified initial state**, $\|e^{Bt}u_0\|_2$, not the maximum over unit states $\|e^{Bt}\|_2$. With initial state $e_1$, B also gives pure $e^{-t}$ decay.

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## The symmetric part tests contraction for every initial state

For a real system u′=Bu,

$$\frac{d}{dt}\|u\|_2^2 = u^T(B+B^T)u = 2u^T H u, \qquad H=(B+B^T)/2.$$

Every initial state has a nonincreasing norm exactly when H is negative semidefinite. Sufficiency follows along each solution. For necessity, evaluate the derivative at t=0 for arbitrary initial vectors. The complex version uses conjugate transpose.

Here $H=\begin{bmatrix}-1 & 2\\ 2 & -1\end{bmatrix}$ has eigenvalues 1,−3. Our $u_0$ is a unit eigenvector for 1, giving initial energy derivative 2. In contrast, the three-dimensional matrix in Exercise 8 has maximum H-eigenvalue $-1+\sqrt{2}/2<0$, so every nonzero solution strictly contracts. **Neither defectiveness nor nonnormality alone proves transient amplification.**

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## Converting a second-order ODE

$$u'' + Bu' + Cu = 0.$$

For scalar constants $B, C$, set $v = u'$.

$$\frac{d}{dt}\begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -C & -B \end{bmatrix}\begin{bmatrix} u \\ v \end{bmatrix}.$$

Characteristic polynomial:

$$\det\begin{bmatrix} -\lambda & 1 \\ -C & -B-\lambda \end{bmatrix} = \lambda^2 + B\lambda + C.$$

Complex conjugate eigenvalues produce oscillation and their real parts determine decay. The initial data $(u(0), u'(0))$ become a two-dimensional initial vector.

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## Real eigenvalues of real symmetric matrices

Let $S \in \mathbb{R}^{n \times n}$, $S = S^T$, and choose a nonzero $x \in \mathbb{C}^n$ with $Sx = \lambda x$. Real entries and symmetry give $S^* = \overline{S}^T = S$. Using the conjugate transpose,

$$x^* S x = \lambda x^* x.$$

because

$$\overline{x^* S x} = x^* S^* x = x^* S x,$$

the numerator is real. Since $x^* x > 0$,

$$\lambda = \frac{x^* S x}{x^* x} \in \mathbb{R}.$$

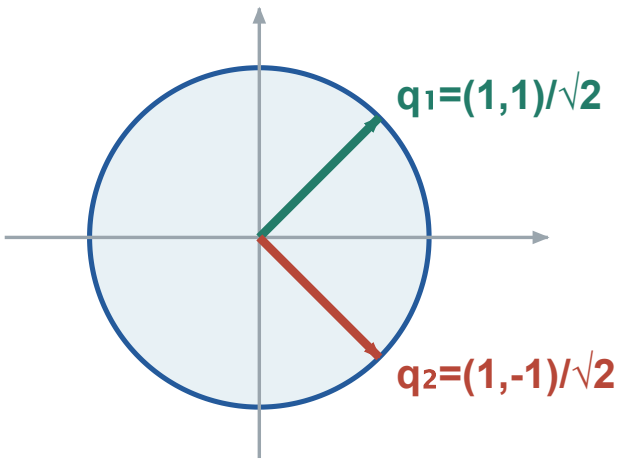
The first equality uses conjugation of a scalar; the second uses $S^* = S$, which does not follow from $S^T = S$ for a complex matrix. The denominator $x^* x = \sum_j |x_j|^2 > 0$ establishes the conclusion.

The counterexample $S = [i]$ satisfies $S^T = S$ but $S^* = [-i] \neq S$, with eigenvalue $i$. The complex-space premise is Hermitian symmetry $S^* = S$. The same proof gives real eigenvalues, and diagonalization uses a unitary matrix $S = U \Lambda U^*$.

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# Real eigenvalues of real symmetric matrices (continued)

Orthonormal eigenbasis: $Q^TSQ=\text{diag}(4,2)$




**$S=[[3,1],[1,3]]$**

**$Q=(1/\sqrt{2})[[1,1],[1,-1]]$**

$Q^TSQ=\text{diag}(4,2)$

$S=Q\Lambda Q^T$; the axes are perpendicular.

Normalize (1,1),(1,-1) before forming Q.

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# Real eigenvalues of real symmetric matrices (continued)

## Complex symmetric does not imply Hermitian

**$S=[i]$ · $S^T=S$**

Only eigenvalue: i

$S^*=[-i]\neq S$

**The Hermitian premise is necessary.**

**Hermitian：$S^*=S$**

$x^*Sx=\lambda x^*x$

$x^*x>0 \Rightarrow \lambda\in R$

**$S=U\Lambda U^*$, with U unitary**

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## Orthogonality of distinct eigenspaces

Let $S \in \mathbb{R}^{n \times n}$ be real symmetric, with real eigenvectors $x, y$. If $Sx = \lambda x$, $Sy = \mu y$,

$$\lambda x^T y = (Sx)^T y = x^T S^T y = \mu x^T y.$$

Consequently,

$$(\lambda - \mu) x^T y = 0.$$

then $\lambda \neq \mu$ implies $x \perp y$. A complete orthonormal eigenbasis follows by induction. The continuous quadratic form $x^T S x$ attains a maximum on the unit sphere; its extremum condition gives a real unit eigenvector $q$. For $v \perp q$, $q^T S v = (Sq)^T v = 0$, so $q^\perp$ is invariant. The restriction remains symmetric. Induction gives a complete orthonormal eigenbasis; within repeated eigenspaces, use Gram–Schmidt.

Spectral theorem：

$$S = Q \Lambda Q^T.$$

For example,

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## Orthogonality of distinct eigenspaces (continued)

$$S = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}, \qquad Q = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

Since $Q^T Q = I$ and

$$Q^T S Q = \begin{bmatrix} 4 & 0 \\ 0 & 2 \end{bmatrix},$$

we obtain $S = Q \Lambda Q^T$. The unnormalized vectors $(1, 1)^T, (1, -1)^T$ specify directions only. Divide them by $\sqrt{2}$ before using them as columns of $Q$.

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## Equivalent positive-definiteness conditions

For real symmetric $S$, the following conditions are equivalent.

1. $x^T S x > 0$ for every $x \neq 0$.
2. Every eigenvalue is positive.
3. Every leading principal minor is positive.
4. Symmetric elimination without swaps has positive pivots.
5. $S = A^T A$ for some full-column-rank $A$.
6. $S = RR^T$ for an invertible lower-triangular $R$ with positive diagonal, the Cholesky factorization.

Symmetric elimination completes squares, giving $S = LDL^T$ and $x^T S x = (L^T x)^T D (L^T x)$. Positive pivots are therefore equivalent to positive energy. Leading minors satisfy $\Delta_k = d_1 \cdots d_k$, so positivity of all leading minors is equivalent to positive pivots. Taking $R = LD^{1/2}$ gives Cholesky, and $A = R^T$ gives the Gram form. Conversely, squared lengths give positive energy. A positive determinant alone is insufficient; $-I_2$ has determinant 1 but is negative definite.

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# Equivalent positive-definiteness conditions (continued)

$2 \times 2$ :

$$S = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$$

is positive definite exactly when $a > 0, ac - b^2 > 0$.

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## Energy proof and counterexamples

$$x^T S x = \sum_i \lambda_i (q_i^T x)^2.$$

If all $\lambda_i > 0$, every nonzero $x$ has some $q_i^T x \neq 0$, making the energy positive. Conversely, an eigenvalue $\lambda_i \leq 0$ supplies a nonzero eigenvector with nonpositive energy, contradicting strict positive definiteness.

For a counterexample,

$$S_1 = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$$

eigenvalues $3, -1$，not positive definite。

If $A$ has independent columns,

$$x^T A^T A x = \|Ax\|^2 > 0$$

for every $x \neq 0$, so $A^T A$ is positive definite. Full row rank cannot replace column independence.

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## Positive semidefinite boundary

$$S = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}.$$

At $x = (1, -1)^T$, $x^T S x = 0$; the eigenvalues are $2, 0$. Thus the matrix is positive semidefinite but not positive definite.

For $S = A^T A$, dependent columns give nonzero $x$ with $Ax = 0$ and zero energy. Conversely, zero energy forces $Ax = 0$.

A semidefinite matrix can have an $LDL^T$ factorization, but Cholesky must handle zero pivots. A pseudoinverse does not make a semidefinite matrix invertible.

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## Ellipsoids and conditioning

Positive definite $S$:

$$x^T S x = 1$$

is an origin-centered ellipsoid. In eigen-coordinates,

$$y = Q^T x,$$
$$\sum_i \lambda_i y_i^2 = 1.$$

The semiaxis length in direction $i$ is

$$1/\sqrt{\lambda_i}.$$

The smallest eigenvalue gives the longest semiaxis. Also,

$$\kappa_2(S) = \frac{\lambda_{\max}}{\lambda_{\min}}.$$

Positive definiteness guarantees exact invertibility, but a small minimum eigenvalue relative to the largest still gives poor conditioning.

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## Chapter check

You can now do the following.

- Find eigenvalues from the characteristic equation.
- Distinguish algebraic and geometric multiplicities.
- Give defective counterexamples.
- Solve powers and ODEs by diagonalization when an eigenbasis exists.
- Combine complex conjugate modes into real solutions.
- Prove the real symmetric spectral theorem and orthogonality relations.
- Connect energy, minors, pivots, Cholesky, and Gram factorizations.
- Interpret conditioning through ellipsoid semiaxes.

SVD in Chapter 7 applies to rectangular matrices and does not require a complete eigenbasis of the original matrix.

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# References and attribution

Gilbert Strang. Introduction to Linear Algebra, 5th Edition, 2016.

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